In-domain and Cross-domain Transferability for Damage Detection in Structural Health Monitoring

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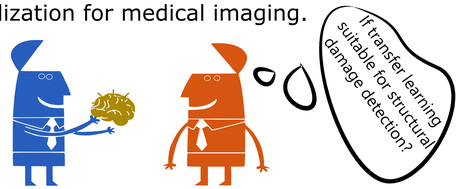
Introduction

Large amount of images data is collected in visual inspection of civil structures. Manual damage identification from such a massive data source demands tremendous efforts and is prone to discrepancies due to human errors, fatigue, poor judgments, and lack of experience of inspectors.

Transfer learning is a natural solution for learning and improving the performance of such tasks having small labeled datasets as shown for several vision tasks. However, recent research reported scepticism in generalibility of transfer learning for specfic domains.

For example

- He et al. [7] found that transfer learning may not result in performance improvements even when the source & target tasks are of similar nature.
- Raghu et al. [8] proved that transfer offers little benefit compared to the random initialization for medical imaging.



Therefore, the objective of this study is to investigate the potential advantages and downsides of in-domain transfer learning for structural damage detection using visual data and provide a comparison to the state-of-the-art cross-domain transfer.

Datasets

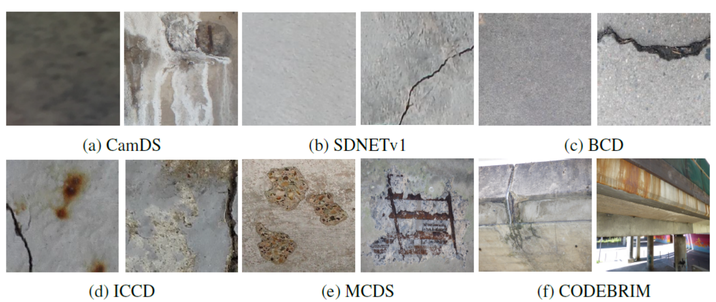
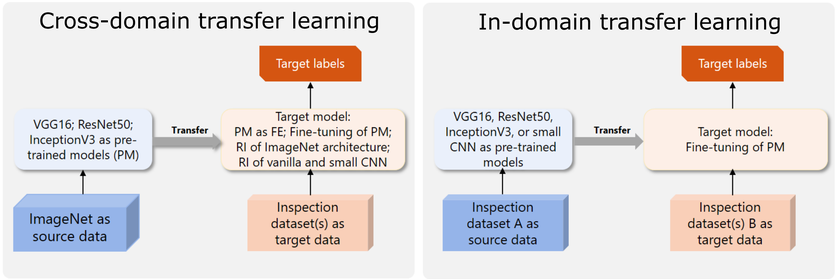


Figure 1: Examples images from datasets. From left to right: (a) Complex lighting with no crack, Unhealthy with crack - efflorescence and exposed bars. (b) No crack, Minor crack. (c) No crack, Large crack on deck. (d) Crack with rust stains, Crack with minor scaling. (e) Concrete scaling, Corrosion with exposed bars. (f) Spallation and efflorescence, Rust stains.

Table 1: Overview of bridge inspection datasets.

Name	Source	Year	Size	Image size	Classes	Problem
CamDS	Huethwohl [2]	2017	1,027	299 × 299	2	Binary
SDNETv1	Maguire et al. [4]	2018	13,620	256 × 256	2	Binary
BCD	Xu et al. [6]	2019	5,390	224 × 224	2	Binary
ICCD	Li and Zhao [3]	2019	60,010	256 × 256	2	Binary
MCDS	Brilakis et al. [1]	2019	2,411	400 × 400	10	Multi-label
CODEBRIM	Mundt et al. [5]	2019	8,304	400 × 400	6	Multi-label

Approach



PM:Pre-trained models, FR: Feature extraction, RI: Random initialization

Experiments and results

Are pre-trained IMAGENET models and features helpful for damage detection tasks compared to vanilla (relatively small) CNN models trained from scratch?

Table 2: Performance comparison of CBR models (small/vanilla CNNs) and standard IMAGENET architectures with different initialization with ROC-AUC score. VGG16 with fine-tuning performed best for five datasets, followed by ResNet, shown with bold entries. Under random initialization and fixed weights settings, the CBR models performed better than IMAGENET model for four datasets followed by VGG16, as shown with underlined entries.

		CamDS	SDNETv1	BCD	ICCD	MCDS	CODEBRIM
Random Init.	CBR Tiny	0.79±0.02	0.74±0.01	0.97±0.00	0.93±0.13	0.67±0.02	0.80±0.04
	CBR Small	0.78±0.02	0.72±0.01	0.96±0.0	0.98±0.0	0.67±0.02	0.80±0.03
	CBR LargeW	0.75±0.03	0.74±0.01	0.97±0.01	0.96±0.02	0.67±0.01	0.80±0.04
	CBR LargeT	0.74±0.02	0.67±0.0	0.96±0.00	0.96±0.02	0.64±0.01	0.79±0.01
	InceptionV3	0.69±0.03	0.60±0.06	0.90±0.03	0.96±0.01	0.61±0.01	0.77±0.02
	VGG16	0.78±0.04	0.50±0.00	0.98±0.01	0.78±0.26	0.59±0.01	0.82±0.01
	ResNet50	0.69±0.02	0.61±0.08	0.60±0.20	0.93±0.03	0.57±0.01	0.73±0.02
Transfer (Fine-Tuned)	InceptionV3	0.73±0.02	0.78±0.01	0.98±0.00	0.98±0.0	0.72±0.01	0.89±0.0
	VGG16	**0.82±0.02**	**0.84±0.0**	**0.99±0.0**	**0.98±0.0**	**0.77±0.03**	0.88±0.01
	ResNet50	0.62±0.02	0.82±0.01	0.98±0.01	0.98±0.0	0.54±0.01	**0.90±0.01**
Fixed (Feature Ext.)	InceptionV3	0.55±0.01	0.50±0.01	0.52±0.02	0.67±0.04	0.54±0.0	0.55±0.01
	VGG16	0.77±0.01	0.72±0.02	0.98±0.01	0.93±0.00	0.64±0.01	0.78±0.00
	ResNet50	0.50±0.00	0.50±0.00	0.50±0.00	0.50±0.01	0.50±0.00	0.50±0.00

Are domain-specific representations useful with or without pre-trained IMAGENET features for new and previously unseen tasks?

Table 3: ROC-AUC performance scores for in-domain transfer per dataset. The rows represents the upstream source model trained for target task and the column shows the results of in-domain transfer for each dataset. The bold entries depict the best performing models compared to the random initialization

	CamDS	SDNETv1	BCD	ICCD	MCDS	CODEBRIM
Rand. Init.	**0.79±0.02**	0.74±0.01	0.98±0.01	0.97±0.0	0.67±0.02	**0.82±0.01**
CamDS		0.73±0.05	0.99±0.00	0.96±0.01	0.67±0.02	0.80±0.03
SDNETv1	0.75±0.01		0.98±0.00	0.97±0.00	0.67±0.01	0.77±0.03
BCD	0.77±0.04	**0.81±0.00**		**0.98±0.00**	0.61±0.01	0.81±0.00
ICCD	0.78±0.02	0.80±0.00	**1.00±0.00**		0.69±0.00	0.76±0.03
MCDS	0.76±0.02	0.72±0.07	0.98±0.00	0.97±0.01		0.79±0.02
CODEBRIM	0.75±0.04	0.80±0.01	0.98±0.00	0.98±0.00	**0.70±0.01**	

How well do in-domain and cross-domain representations perform with small sets of labeled training samples?

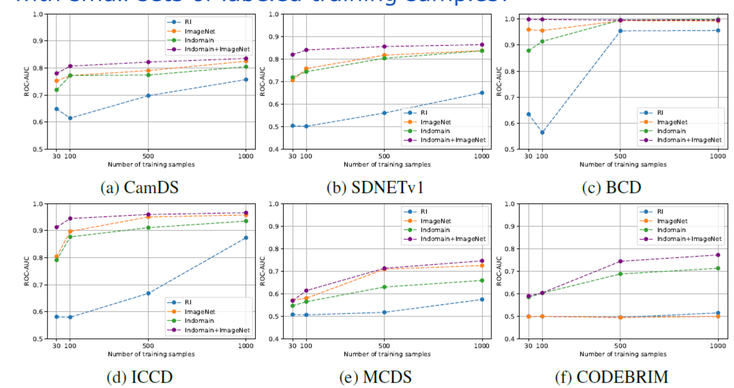


Figure 2: ROC-AUC score on test-set after training with different initialization over a limited number of training samples from each dataset.

Conclusions

Transfer learning using pre-trained ImageNet models has shown significant performance gains for the structural damage detection task. Additionally, in-domain transfer is a practically useful approach, when training efficiency and flexibility in the models' architecture is desirable. Finally, combination of representations from in-domain & ImageNet have consistently shown superior performance, even with very small labeled data, at the cost of extensive fine-tuning of large pre-trained models.

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